



Predictive Modeling and Analysis of Airbnb Listing Prices Using Machine Learning

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1 Introduction

1.1 Background

The rapid growth of online platforms has transformed the way people search for and book accommodation. Airbnb, as one of the leading platforms in the sharing economy, provides a wide variety of listings with diverse pricing structures influenced by multiple factors. Understanding how these factors affect pricing is essential for both hosts and customers, as it can support better pricing strategies and decision-making. With the increasing availability of data, machine learning techniques offer powerful tools to analyze complex relationships and uncover patterns that traditional methods may not capture effectively.

1.2 Dataset Description

The dataset used in this study is derived from publicly available Airbnb listing data provided by Inside Airbnb, which offers detailed, real-world data collected from the Airbnb platform. The dataset contains comprehensive information about listings in New York City, including location coordinates (latitude and longitude), room type, price, minimum nights, number of reviews, and availability throughout the year. In addition to the original features, several engineered variables were created to enhance the analysis, such as demand score, availability ratio, and distance to the city center. This dataset provides a rich representation of listing characteristics and user behavior, making it suitable for both predictive modeling and clustering analysis.

1.3 Aim and Objectives

The main aim of this project is to analyze and predict Airbnb listing prices using machine learning techniques. To achieve this goal, several objectives were defined. First, to clean and preprocess the dataset to ensure data quality and consistency. Second, to perform exploratory data analysis to understand the relationships between variables and identify key patterns. Third, to build and evaluate multiple regression models in order to determine the most effective approach for price prediction. Finally, to apply clustering techniques to identify natural groupings within the data and gain additional insights into listing behavior.

1.4 Expected Outcomes

The expected outcome of this project is to develop a reliable model capable of accurately predicting Airbnb listing prices based on available features. It is also anticipated that the analysis will highlight the most important factors influencing pricing, particularly those related to location and property characteristics. Additionally, clustering is expected to reveal meaningful segmentation of listings based on usage patterns, availability, and demand. Overall, the project aims to provide both predictive insights and a deeper understanding of the factors that drive pricing in the Airbnb marketplace.

2 Data Cleaning and Feature Engineering

2.1 Data Cleaning

Data cleaning is a crucial step in preparing the dataset for analysis and modeling, as it ensures accuracy, consistency, and reliability of the data. The raw dataset contained inconsistencies such as invalid values, missing entries, and irrelevant features that could negatively affect model performance. Therefore, several preprocessing steps were applied to improve data quality and prepare it for further analysis.

2.1.1 Handling Invalid Values

During the initial inspection of the dataset, it was observed that certain numerical features contained invalid values, particularly zeros in variables such as price and minimum nights. These values are unrealistic in the context of Airbnb listings and may indicate missing or incorrect data entries. To address this issue, such values were replaced with missing values (NaN) to allow for appropriate handling in later stages. This step ensured that invalid entries did not distort statistical analysis or model training.

2.1.2 Missing Value Analysis

A detailed analysis of missing values was conducted to understand their distribution and significance. It was found that the variable representing reviews per month contained missing values, primarily for listings with no reviews. Since the absence of reviews logically corresponds to zero activity, these missing values were replaced with zero. For other variables, such as price, rows containing missing values were removed to maintain data integrity, as imputing such critical values could introduce bias. This approach ensured a balance between preserving data and maintaining accuracy.

2.1.3 Feature Transformation

Several feature transformations were applied to enhance the dataset and improve model performance. The date of the last review was converted into a numerical feature representing the number of days since the last review, allowing the model to capture recency effects. Missing values in this feature were replaced with a default value to indicate the absence of reviews. Additionally, the target variable, price, was transformed using a logarithmic function to reduce skewness and stabilize variance. This transformation improves the performance of regression models by making the distribution more symmetric and easier to model.

2.1.4 Dropping Irrelevant Features

Certain features in the dataset were identified as irrelevant or non-informative for predictive modeling. For example, textual identifiers such as listing names and host names do not contribute meaningful numerical information and may introduce unnecessary noise. These features were removed from the dataset to simplify the model and reduce dimensionality. This step helps improve computational efficiency and ensures that the model focuses only on relevant and informative variables.

2.2 Feature Engineering

Feature engineering was performed to create new variables that better capture the underlying patterns in the data and improve the predictive performance of the models. By transforming and combining existing features, additional meaningful information was extracted, allowing the models to better understand relationships between variables.

2.2.1 Demand-Based Features

To represent the level of demand for each listing, a new feature called demand score was created by combining the number of reviews and reviews per month. This feature captures both the historical popularity of a listing and its recent activity, providing a more comprehensive measure of demand. By integrating these two variables, the model is better able to identify listings that are consistently popular over time.

2.2.2 Availability Features

An availability-related feature was introduced to reflect how often a listing is accessible throughout the year. Specifically, an availability ratio was calculated by dividing the number of available days by the total number of days in a year. This transformation normalizes the availability feature, making it easier to compare across listings. It also helps the model distinguish between listings that are frequently available and those that are rarely accessible.

2.2.3 Host-Based Features

To capture host behavior and professionalism, a new binary feature was created based on the number of listings managed by a host. Listings owned by hosts with more than one property were classified as professional hosts. This feature helps differentiate between individual hosts and those managing multiple listings, which may influence pricing strategies and listing quality. While textual identifiers such as host names were removed due to their lack of predictive value, meaningful host-related information was retained and transformed into structured features. In particular, this binary variable provides a more informative representation of host activity without introducing unnecessary noise into the model.

2.2.4 Geographical Features

Location plays a critical role in determining Airbnb prices. To better represent geographical influence, a new feature measuring the distance of each listing from the city center was created using latitude and longitude coordinates. This feature provides a continuous measure of location, allowing the model to capture spatial relationships more effectively than categorical neighborhood labels alone. It also enables a more precise understanding of how proximity to central areas impacts pricing.

2.2.5 Target Variable Transformation

The target variable, price, exhibited a highly skewed distribution with a long right tail, which can negatively impact model performance. To address this issue, a logarithmic transformation was applied to create a new variable, log price. This transformation reduces skewness, stabilizes variance, and improves the ability of regression models to learn meaningful patterns. As a result, the transformed target variable provides a more suitable representation for predictive modeling.

While standardization techniques such as z-score normalization are useful for scaling input features, they do not address skewness in the target variable. Therefore, a logarithmic transformation was applied alone to the price variable.

3 Exploratory Data Analysis

3.1 Histogram Analysis

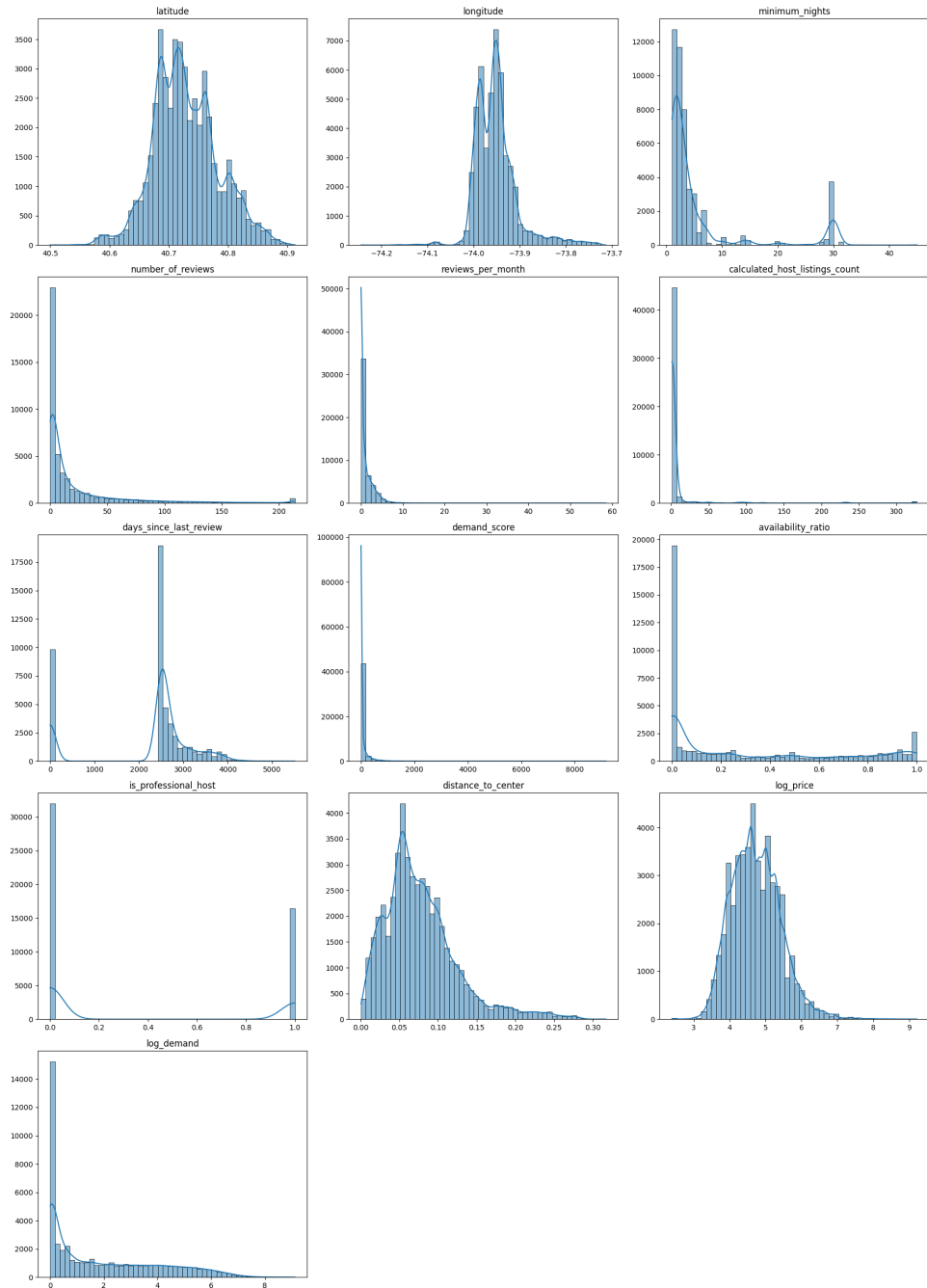


Figure 1: Distribution of numerical features

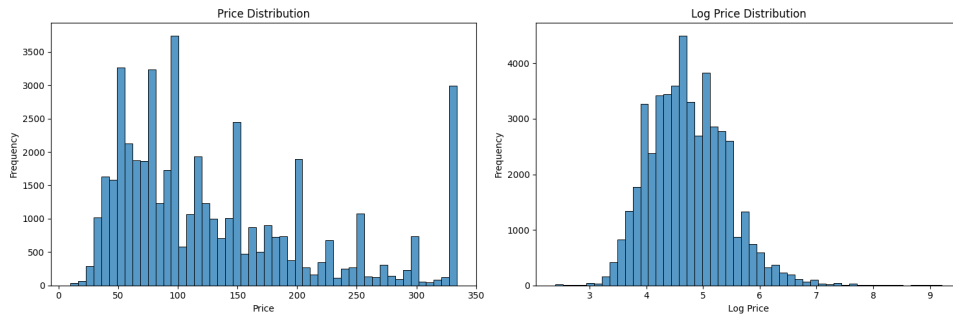


Figure 2: Distribution of Price and log-transformed price

Histogram analysis was performed to examine the distribution of numerical features, particularly the target variable. The original price distribution was highly right-skewed, indicating the presence of a large number of low-priced listings and a small number of very expensive listings. To address this issue, a logarithmic transformation was applied, resulting in a more symmetric distribution of log price. Histograms of other numerical features revealed varying distributions, with some variables showing skewness and concentration around specific values. These observations highlighted the need for transformation and normalization to improve model performance.

3.2 Boxplot Analysis

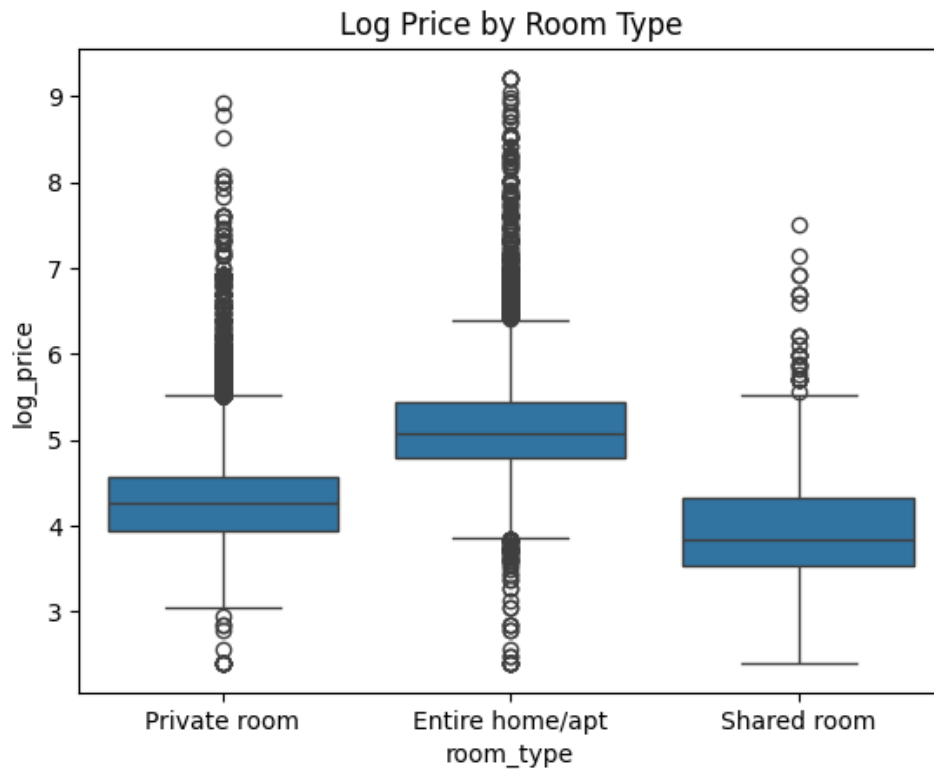


Figure 3: Boxplot showing Log Price by Room Type

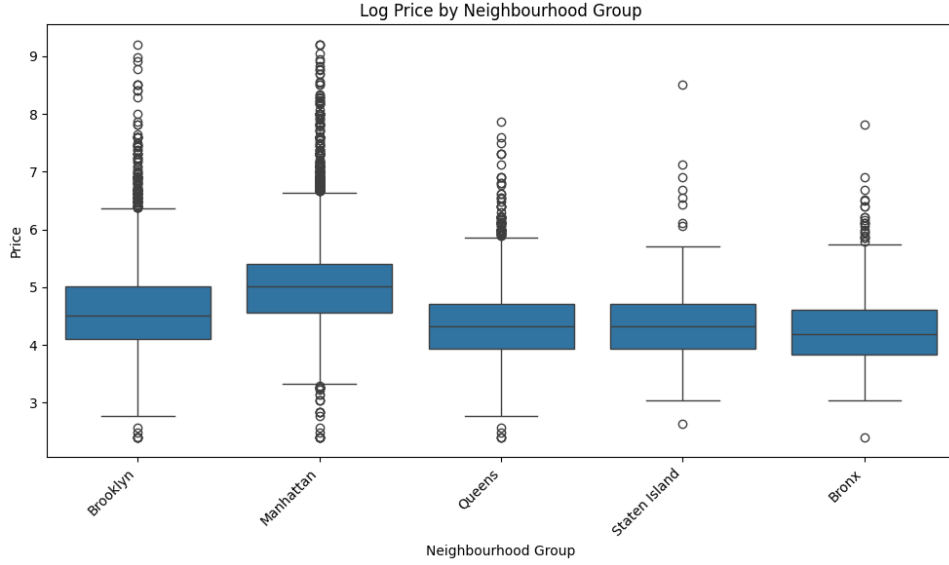


Figure 4: Boxplot showing Log Price by Neighbourhood Groups

Boxplots were used to analyze the relationship between categorical variables and price. The analysis showed that room type has a significant impact on pricing, with entire homes or apartments generally having higher prices compared to private or shared rooms. Additionally, the neighbourhood group also influenced price variation, with certain areas exhibiting higher median prices than others. The presence of outliers in both plots indicates that some listings are priced significantly higher than the majority, reflecting the diversity of the Airbnb market.

3.3 Scatter Plot Analysis

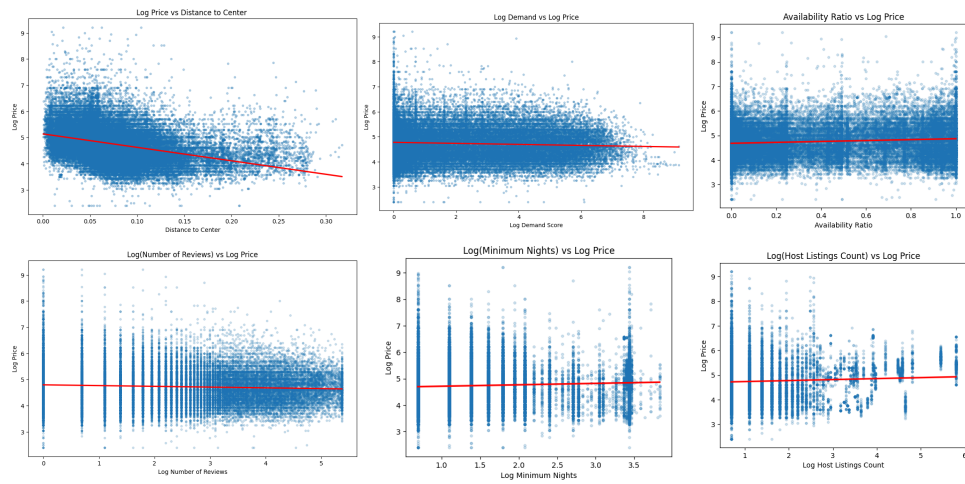


Figure 5: Relationship between numerical features and log price

Scatter plots were used to explore the relationships between numerical features and log price. The analysis revealed that distance to the city center has a noticeable inverse relationship with price, where listings closer to the center tend to be more expensive. Demand-related features, such as the number of reviews and demand score, showed weak relationships with price, suggesting that popularity does not strongly determine pricing. Similarly, features such as minimum nights and availability exhibited limited correlation with price, indicating that pricing is influenced more by structural and location-based

factors than by user engagement or booking patterns. Overall, the scatter plots confirmed the presence of non-linear relationships in the data.

3.4 Correlation Heatmap Analysis

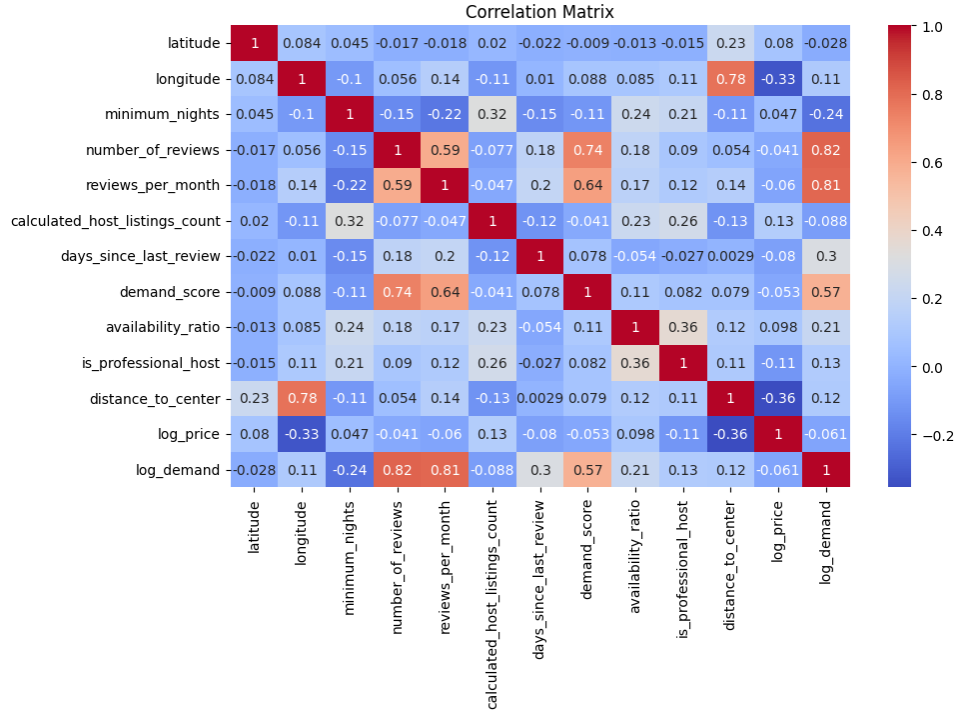


Figure 6: Correlation matrix of numerical features

A correlation heatmap was used to quantify the relationships between numerical variables. The results showed that most features have weak to moderate correlations with each other, indicating low multicollinearity within the dataset. The strongest relationships were observed among engineered features, such as demand score and number of reviews, which are naturally related. However, no single feature exhibited a strong linear correlation with price, suggesting that the relationship between features and price is complex and likely non-linear. This finding supports the use of advanced machine learning models capable of capturing such relationships.

3.5 Skewness and Kurtosis Analysis

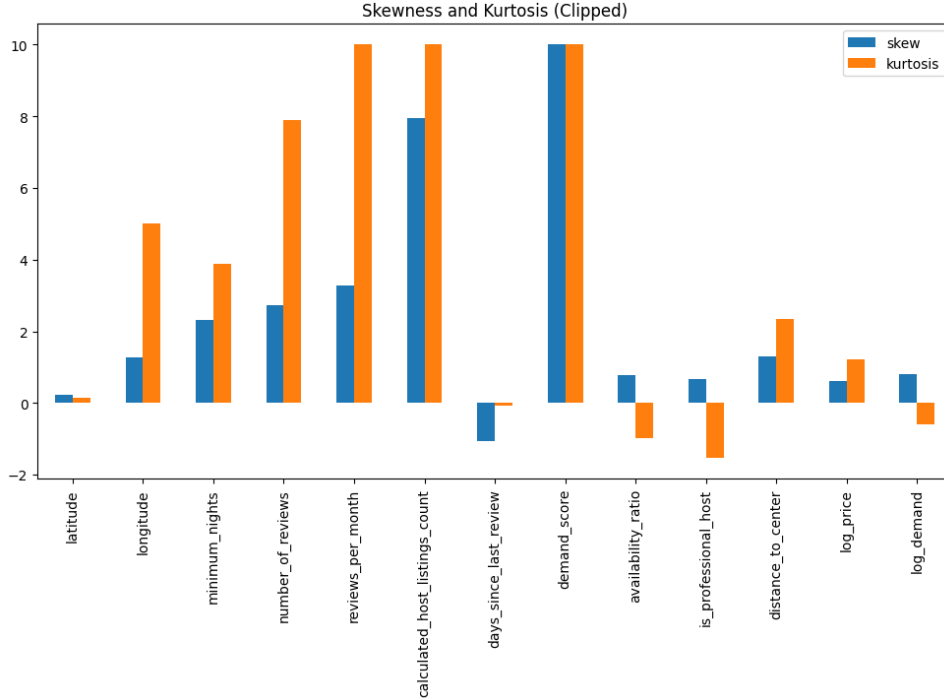


Figure 7: Skewness and kurtosis distribution of numerical features

Skewness and kurtosis analysis was conducted to further examine the distribution characteristics of numerical features. Several variables exhibited positive skewness, indicating long right tails and the presence of extreme values. The application of log transformation to the price variable significantly reduced its skewness, making it more suitable for modeling. Kurtosis analysis revealed that some features contain heavy tails, suggesting the presence of outliers. These findings emphasize the importance of transformation and robust modeling techniques to handle non-normal distributions and extreme values effectively.

4 Modeling

4.1 Problem Definition

The objective of this stage is to develop machine learning models capable of predicting Airbnb listing prices based on available features. Since the target variable is continuous, the problem is formulated as a regression task. In addition, clustering is applied as an unsupervised learning approach to identify hidden patterns and segment listings based on their characteristics, providing complementary insights beyond prediction.

To improve model performance and address skewness in the data, a logarithmic transformation was applied to the price variable, resulting in the use of log price as the target. The goal is to identify the most accurate model while understanding the underlying relationships between features and price.

4.2 Data Preparation

Before training the models, the dataset was preprocessed to ensure consistency and suitability for machine learning algorithms. Numerical features were scaled to standardize their ranges, while categorical variables such as room type and neighbourhood group were encoded using one-hot encoding. The dataset was then split into training and testing sets to evaluate model performance on unseen data. This preprocessing step ensures that the models can learn effectively and generalize well to new data. In addition, a cross-validation strategy was incorporated during model training to ensure robust evaluation and improve generalization performance.

4.3 Model Selection

Three regression models were selected to compare different learning approaches: Linear Regression, Decision Tree Regressor, and Random Forest Regressor. These models were chosen to represent both simple linear methods and more complex non-linear and ensemble techniques.

4.3.1 Linear Regression

Linear Regression was used as a baseline model due to its simplicity and interpretability. It assumes a linear relationship between input features and the target variable. Although easy to implement, its performance is limited when dealing with complex, non-linear relationships in the data.

4.3.2 Decision Tree Regressor

The Decision Tree Regressor was selected for its ability to model non-linear relationships by recursively splitting the dataset based on feature values. It provides more flexibility than linear models but can be prone to overfitting if not properly controlled.

4.3.3 Random Forest Regressor

The Random Forest Regressor is an ensemble method that combines multiple decision trees to improve prediction accuracy and reduce overfitting. By averaging the outputs of several trees, it produces more stable and reliable predictions, making it well-suited for complex datasets.

4.4 Model Training

All models were trained using the processed training dataset. To ensure a fair comparison, the same training and testing split was applied across all models.

A 5-fold cross-validation strategy was used during training to evaluate model performance and improve generalization. This approach reduces the risk of overfitting by validating the model on multiple subsets of the data.

For the Decision Tree and Random Forest models, hyperparameter tuning was performed using GridSearchCV. The Decision Tree model was optimized by adjusting parameters such as maximum depth and minimum samples split to control model complexity and prevent overfitting. Similarly, the Random Forest model was tuned by varying the number of estimators, maximum depth, and minimum samples split to improve predictive performance and stability.

A reduced hyperparameter search space was used to ensure computational efficiency while still exploring meaningful parameter combinations. The best-performing models identified through cross-validation were then evaluated on the test dataset.

4.5 Model Evaluation

The models were evaluated using three key metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score. These metrics provide insight into prediction accuracy and model performance. The evaluation results are based on models optimized through cross-validation and hyperparameter tuning, ensuring a reliable comparison of model performance.

Table 1: Model Performance Comparison

Model	MAE	RMSE	R^2
Linear Regression	0.348	0.482	0.518
Decision Tree	0.332	0.467	0.547
Random Forest	0.314	0.442	0.595

The results show that Random Forest achieved the best performance, with the lowest error values and the highest R^2 score, followed by the Decision Tree model, while Linear Regression performed the worst.

4.6 Prediction Analysis

To better understand model behavior, predicted values from the trained models were compared with actual values. The results indicate that most predictions closely follow the diagonal line, suggesting good overall performance. However, the model tends to underestimate higher-priced listings and slightly overestimate lower-priced ones, indicating a regression-to-the-mean effect.

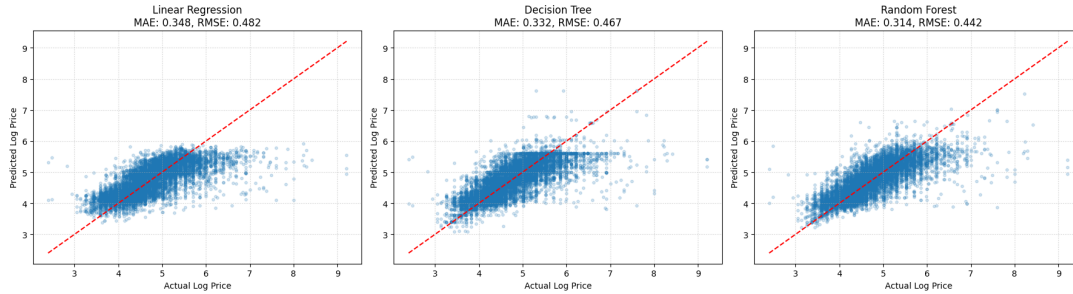


Figure 8: Actual vs Predicted Values for All Models

To further examine the performance of the best model, the tuned Random Forest model predictions were analyzed separately. The plot shows that predictions are closely aligned with the diagonal line, confirming the model's strong performance. However, some deviation is observed for higher-priced listings, indicating that extreme values remain challenging to predict accurately.

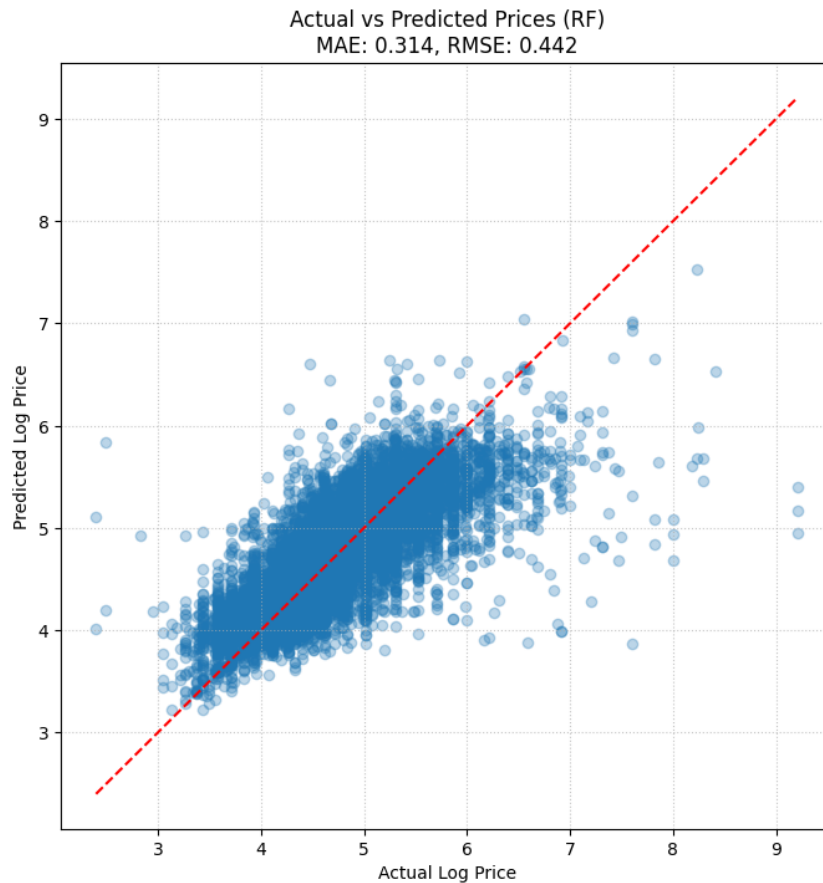


Figure 9: Random Forest Actual vs Predicted Plot

To provide a numerical illustration of the tuned model's predictions, a sample of actual and predicted values from the Random Forest model is presented in Table 2.

Table 2: Sample of Actual vs Predicted Values (Random Forest Model)

Index	Actual	Predicted
24402	4.7095	4.7968
44153	4.1109	3.8464
20551	7.8598	5.6463
11985	4.8363	5.3984
15728	3.9318	4.0349
13199	4.1109	3.9336
1201	6.6859	5.6081
9775	5.0173	5.1038
45425	4.4543	4.8242
47489	4.3307	4.3726

4.7 Evaluation Plots

To further assess model performance, the distribution of residuals was analyzed. Residuals represent the difference between actual and predicted values of the tuned Random Forest model and provide insight into the model's error behavior. The residual distribution shows that most errors are concentrated around zero, indicating that the model produces generally accurate predictions with no significant overall bias. The shape of the distribution is approximately symmetric, suggesting consistent performance across most observations. However, a slight skewness is observed, indicating that the model occasionally underestimates higher-priced listings. Additionally, the presence of a few extreme residuals highlights the difficulty of accurately predicting outliers and unusual cases in the dataset.

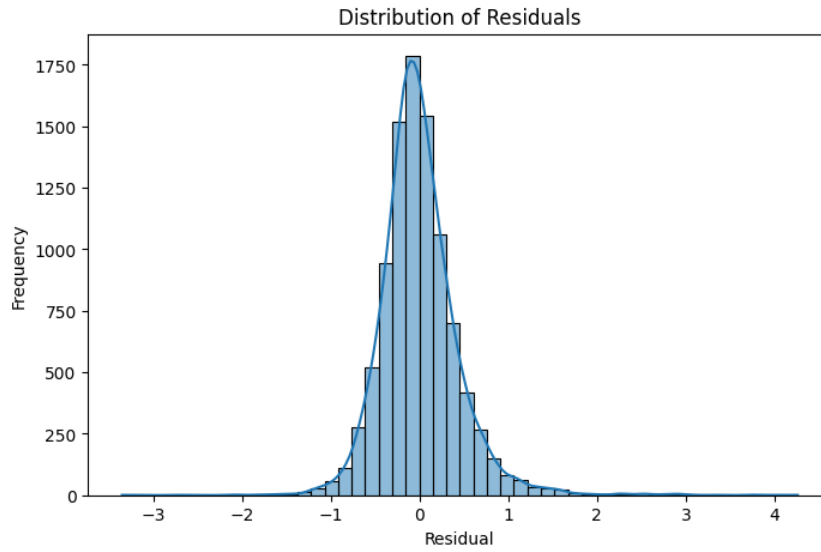


Figure 10: Residual Distribution

Furthermore, the error versus prediction plot demonstrates that prediction errors increase for larger values, highlighting the model's limitations in handling extreme cases.

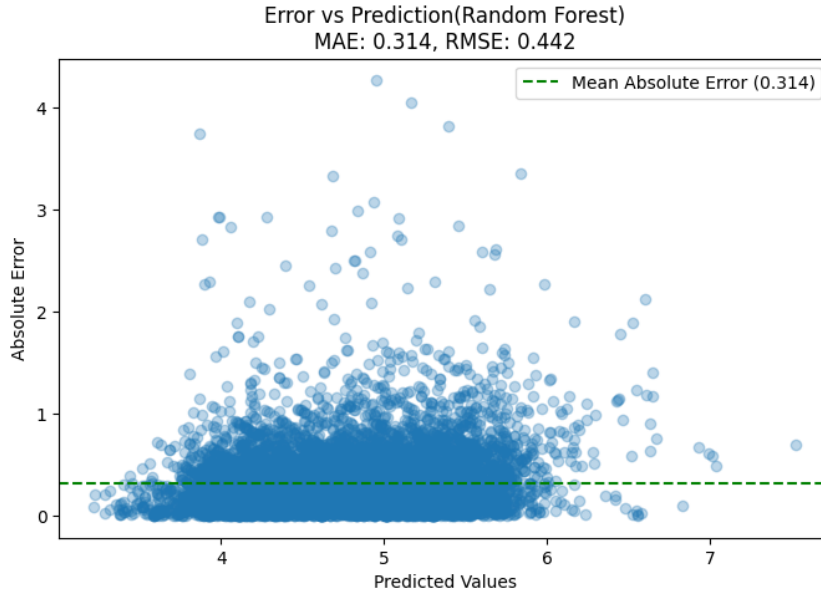


Figure 11: Error vs Prediction Plot

4.8 Feature Importance

Feature importance analysis was conducted using the tuned Random Forest model obtained after hyperparameter optimization to identify the most influential predictors of price. The results show that room type is the most important feature, followed by distance to the city center and geographical coordinates. In contrast, demand-related variables such as number of reviews and availability have relatively lower importance. These findings confirm that pricing is primarily driven by location and structural characteristics rather than user engagement.

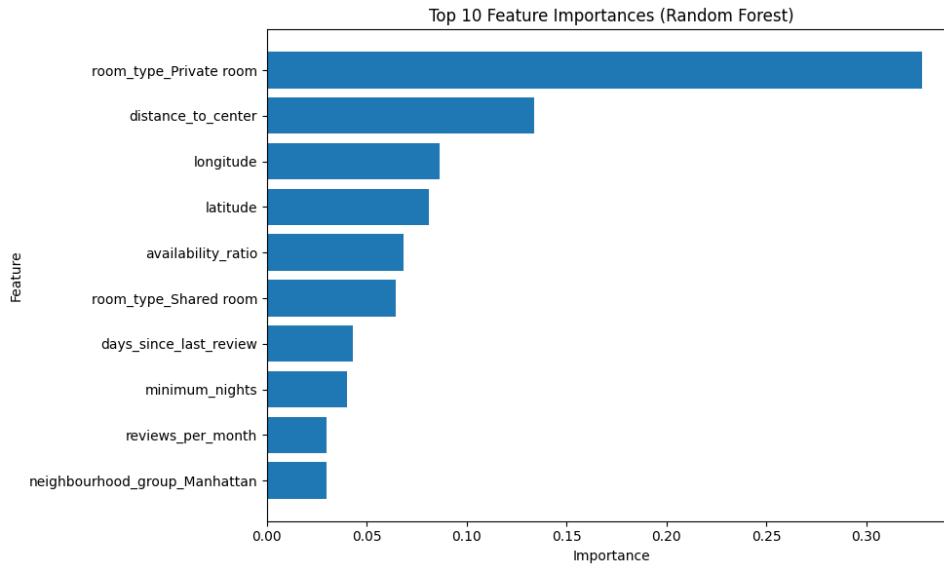


Figure 12: Feature Importance from Random Forest Model

5 K-Means Clustering

5.1 Objective

K-Means clustering was applied as a complementary analysis to identify natural groupings within the Airbnb listings dataset and gain deeper insight into listing characteristics. While regression models focus

on predicting listing prices, clustering is an unsupervised learning technique that does not rely on a target variable but instead groups listings based on similarity across multiple features. The purpose of this analysis is to uncover hidden patterns in listing behavior, pricing structure, and availability, and to better understand how different types of listings can be segmented beyond what can be observed through prediction alone.

5.2 Feature Selection and Preprocessing

A subset of relevant numerical features was selected for clustering, including log price, distance to the city center, availability ratio, minimum nights, and number of reviews. These features were chosen because they capture key aspects of pricing, demand, and listing behavior. Before applying the clustering algorithm, all features were standardized to ensure that they contribute equally to the clustering process. This step is essential because K-Means is sensitive to feature scales and could otherwise be biased toward variables with larger magnitudes.

5.3 Determining the Number of Clusters

To identify the optimal number of clusters, the Elbow Method was used. This method evaluates the within-cluster sum of squares (inertia) for different values of k and identifies the point where the rate of decrease slows significantly.

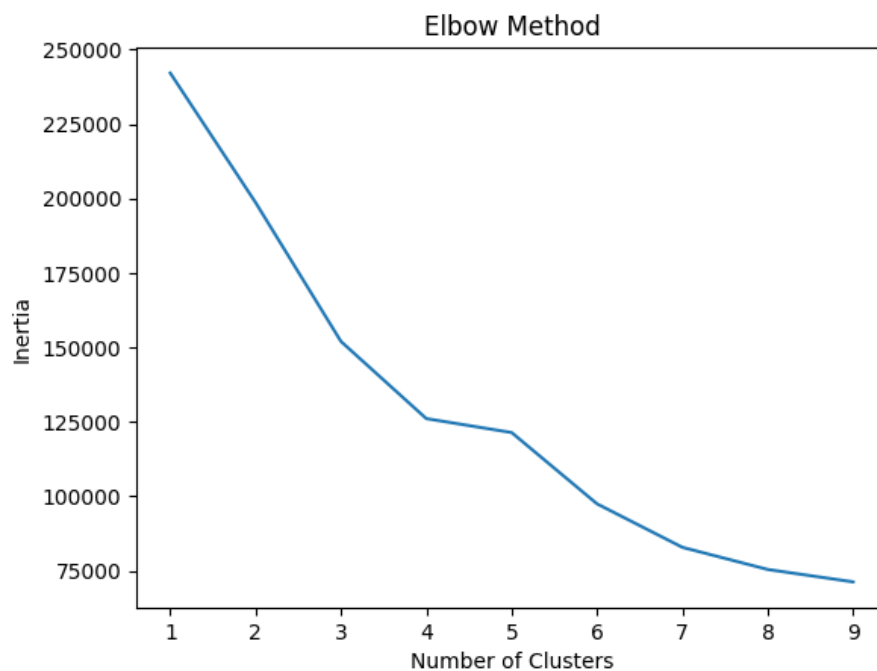


Figure 13: Elbow Method for Optimal Number of Clusters

The plot shows a clear reduction in inertia up to three clusters, after which the improvement becomes more gradual. Based on this observation, the number of clusters was set to three, as it provides a good balance between model simplicity and clustering quality.

5.4 Clustering Results

After applying the K-Means algorithm with three clusters, the dataset was divided into distinct groups representing different types of listings. The summary statistics of each cluster reveal clear differences in pricing, availability, and booking behavior.

Table 3: Cluster Mean Characteristics

Feature	Cluster 0	Cluster 1	Cluster 2
Log Price	4.743	4.689	4.833
Distance to Center	0.068	0.102	0.065
Availability Ratio	0.079	0.666	0.600
Minimum Nights	3.22	2.46	28.92
Number of Reviews	10.09	53.87	7.92

The results indicate that each cluster captures a unique combination of features, highlighting variations in how listings are priced and utilized across the dataset.

5.5 Visualization

To better understand the clustering structure, a scatter plot was used to visualize the clusters based on distance to the city center and log price.

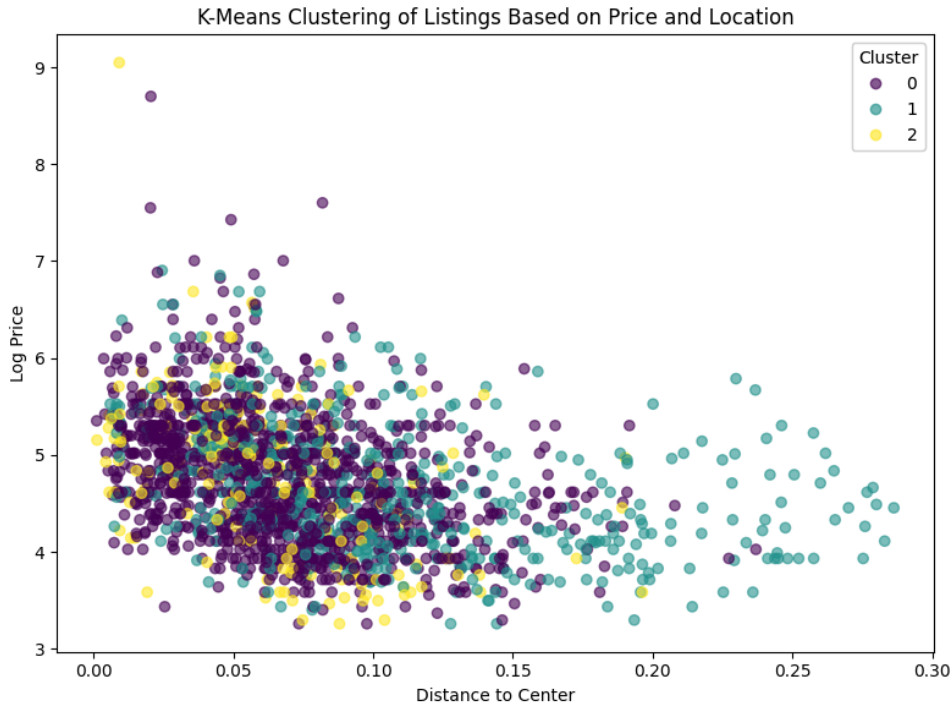


Figure 14: Cluster Visualization (Log Price vs Distance to City Center)

The visualization shows that listings farther from the city center tend to form a more distinct group, while listings closer to the center exhibit more overlap. This overlap is expected because clustering was performed using multiple features, while the visualization represents only two dimensions.

5.6 Cluster Interpretation

The clusters can be interpreted based on their average characteristics. The first cluster represents listings with low availability and moderate pricing, suggesting properties that are rarely available or selectively rented. The second cluster consists of highly active listings with a large number of reviews and high availability, indicating strong demand and frequent bookings. The third cluster is characterized by higher prices and significantly higher minimum stay requirements, representing long-term or premium rental listings.

Overall, the clustering results reveal meaningful segmentation within the dataset, demonstrating that Airbnb listings can be grouped based on usage patterns, availability, and pricing behavior. These insights complement the regression analysis by providing a deeper understanding of the structure of the market.

6 Discussion and Conclusion

6.1 Discussion

The results of this study demonstrate that Airbnb listing prices are influenced by a combination of location, property characteristics, and availability patterns. The exploratory data analysis revealed that price distributions are highly skewed and that location-based features, such as distance to the city center, play a significant role in determining price. Scatter plot analysis further showed that many relationships between features and price are non-linear, which justified the use of more advanced machine learning models.

The modeling results confirmed these observations. Among the tested models, the Random Forest Regressor achieved the best performance, indicating its ability to capture complex, non-linear relationships within the data. In contrast, Linear Regression performed worse due to its assumption of linearity, while the Decision Tree model provided moderate performance but was more sensitive to overfitting. The prediction analysis showed that all models tend to underestimate higher-priced listings and slightly overestimate lower-priced ones, reflecting a regression-to-the-mean effect commonly observed in predictive modeling.

The Decision Tree model shows signs of overfitting due to its sensitivity to training data and higher variance, while Linear Regression underfits the data as it fails to capture non-linear relationships. In contrast, the Random Forest model provides a better balance by reducing overfitting through ensemble learning and improving generalization.

Feature importance analysis highlighted that room type and location-related variables are the most influential factors in determining price, while demand-related features such as reviews and availability play a smaller role. This suggests that pricing is primarily driven by structural and geographical factors rather than user engagement metrics.

In addition to predictive modeling, clustering analysis provided further insight into the dataset by identifying distinct groups of listings. The results revealed meaningful segmentation based on availability, demand, and pricing patterns. Although the clusters overlapped in the visualization, this is expected due to the projection of multi-dimensional data into two dimensions. Overall, clustering complemented the regression analysis by offering a different perspective on how listings can be categorized.

These findings can be useful for Airbnb hosts to optimize pricing strategies and for platforms to better understand market segmentation and user behavior.

6.2 Limitations

Despite the valuable insights obtained, this study has several limitations. First, the dataset represents a snapshot of Airbnb listings and does not capture temporal changes such as seasonal pricing variations or market trends over time. Second, some potentially important features, such as amenities, property condition, and host ratings, were not included in the dataset, which may limit the model's ability to fully explain price variation.

Additionally, the models showed reduced accuracy when predicting extreme values, particularly for high-priced listings. This indicates that outliers and rare cases remain challenging to model accurately. The clustering results, while informative, are also sensitive to the choice of features and the number of clusters, which may affect the interpretation of groupings. Finally, although preprocessing steps were applied carefully, certain assumptions, such as replacing missing values with zeros, may introduce minor bias into the analysis.

6.3 Conclusion

In conclusion, this study successfully applied machine learning techniques to analyze and predict Airbnb listing prices. The results demonstrate that Random Forest is the most effective model among those tested, providing the highest accuracy and best overall performance. The analysis confirmed that location and property type are the primary drivers of price, while demand-related factors have a more limited impact.

Furthermore, K-Means clustering revealed meaningful segmentation within the dataset, highlighting different types of listings based on availability, pricing, and usage patterns. These findings provide valuable insights into the structure of the Airbnb market and can support better decision-making for both hosts and analysts.

Overall, the combination of exploratory analysis, predictive modeling, and clustering offers a comprehensive understanding of the factors influencing Airbnb pricing. Future work could focus on incorporating additional features, exploring advanced models, and analyzing temporal trends to further improve prediction accuracy and deepen insights.

This project provided valuable experience in applying machine learning techniques to real-world data, particularly in data preprocessing, model evaluation, and interpreting results.